Other linear representations of persistence. Many other linear representations of persistence have been proposed in the literature, such as e. g., the persistence silhouette (Chazal et al., 2015b), the accumulated persistence function (Biscio and Møller, 2019) and variants of the persistence surface (Chen et al., 2017; Kusano et al., 2016; Reininghaus et al., 2015).

Considering persistence diagrams as discrete measure and their vectorizations as linear representation is an approach that also proven fruitful to study distributions of diagrams Divol and Chazal (2020) and the metric structure of the space of persistence diagrams Divol and Lacombe (2020) - see Sections 5.6 and 6.3.

5.6 Metrics on the space of persistence diagrams

To exploit the topological information and topological features inferred from persistent homology, one needs to be able to compare persistence diagrams, i.e. to endow the space of persistence diagrams with a metric structure. Although several metrics can be considered, the most fundamental one is known as the *bottleneck distance*.

Recall that a persistence diagram is the union of a discrete multi-set in the half-plane above the diagonal Δ and, for technical reasons that will become clear below, of Δ where the point of Δ are counted with infinite multiplicity. A *matching* - see Figure 12 - between two diagrams dgm_1 and dgm_2 is a subset $m \subseteq \mathrm{dgm}_1 \times \mathrm{dgm}_2$ such that every points in $\mathrm{dgm}_1 \setminus \Delta$ and $\mathrm{dgm}_2 \setminus \Delta$ appears exactly once in m. In other words, for any $p \in \mathrm{dgm}_1 \setminus \Delta$, and for any $q \in \mathrm{dgm}_2 \setminus \Delta$, $(\{p\} \times \mathrm{dgm}_2) \cap m$ and $(\mathrm{dgm}_1 \times \{q\}) \cap m$ each contains a single pair. The *Bottleneck distance* between dgm_1 and dgm_2 is then defined by

$$\mathrm{d_b}(\mathrm{dgm}_1, \mathrm{dgm}_2) = \inf_{\text{matching } m} \max_{(p,q)\in m} \|p - q\|_\infty.$$

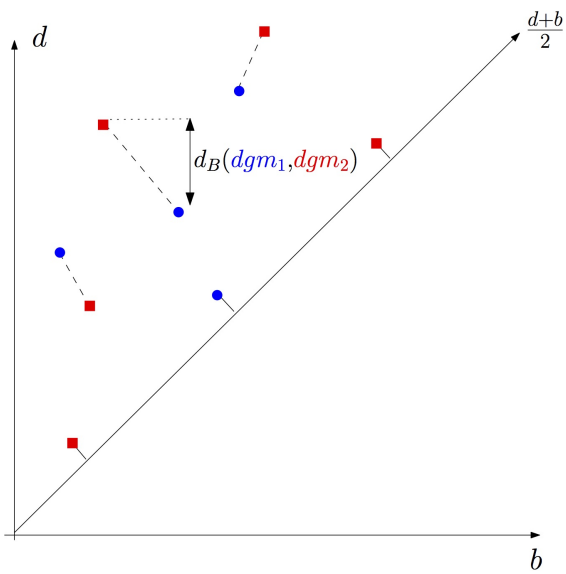

Figure 12: A perfect matching and the Bottleneck distance between a blue and a red diagram. Notice that some points of both diagrams are matched to points of the diagonal.

The practical computation of the bottleneck distance boils down to the computation of a perfect matching in a bipartite graph for which classical algorithms can be used.

The bottleneck metric is a L_∞-like metric. It turns out to be the natural one to express stability properties of persistence diagrams presented in Section 5.7, but it suffers from the same drawbacks as the usual L_∞ norms, i.e. it is completely determined by the largest distance among the pairs and do not take into account the closeness of the remaining pairs of points. A

variant, to overcome this issue, the so-called Wasserstein distance between diagrams is sometimes considered. Given $p \geq 1$, it is defined by

$$W_p(\mathrm{dgm}_1, \mathrm{dgm}_2)^p = \inf_{\text{matching } m} \sum_{(p,q)\in m} \|p-q\|_\infty^p.$$

Useful stability results for persistence in the W_p metric exist among the literature, in particular Cohen-Steiner et al. (2010), but they rely on assumptions that make them consequences of the stability results in the bottleneck metric. A general study of the space of persistence diagrams endowed with W_p metrics has been considered in Divol and Lacombe (2020) where they propose a general framework, based upon optimal partial transport, in which many important properties of persistence diagrams can be proven in a natural way.

5.7 Stability properties of persistence diagrams

A fundamental property of persistence homology is that persistence diagrams of filtrations built on top of data sets turn out to be very stable with respect to some perturbations of the data. To formalize and quantify such stability properties, we first need to precise the notion of perturbation that are allowed.

Rather than working directly with filtrations built on top of data sets, it turns out to be more convenient to define a notion of proximity between persistence module from which we will derive a general stability result for persistent homology. Then, most of the stability results for specific filtrations will appear as a consequence of this general theorem. To avoid technical discussions, from now on we assume, without loss of generality, that the considered persistence modules are indexed by $\mathbb{R}$.

Definition 9. *Let $\mathbb{V}, \mathbb{W}$ be two persistence modules indexed by $\mathbb{R}$. Given $\delta \in \mathbb{R}$, a homomorphism of degree δ between $\mathbb{V}$ and $\mathbb{W}$ is a collection Φ of linear maps $\phi_r : V_r \to W_{r+\delta}$, for all $r \in \mathbb{R}$ such that for any $r \leq s$, $\phi_s \circ v_s^r = w_{s+\delta}^{r+\delta} \circ \phi_r$.*

An important example of homomorphism of degree δ is the *shift endomorphism* $1_{\mathbb{V}}^{\delta}$ which consists of the families of linear maps $(v_{r+\delta}^r)$. Notice also that homomorphisms of modules can naturally be composed: the composition of a homomorphism Ψ of degree δ between $\mathbb{U}$ and $\mathbb{V}$ and a homomorphism Φ of degree δ' between $\mathbb{V}$ and $\mathbb{W}$ naturally gives rise to a homomorphism $\Phi\Psi$ of degree $\delta + \delta'$ between $\mathbb{U}$ and $\mathbb{W}$.

Definition 10. *Let $\delta \geq 0$. Two persistence modules $\mathbb{V}, \mathbb{W}$ are δ-interleaved if there exists two homomorphism of degree δ, Φ, from $\mathbb{V}$ to $\mathbb{W}$ and Ψ, from $\mathbb{W}$ to $\mathbb{V}$ such that $\Psi\Phi = 1_{\mathbb{V}}^{2\delta}$ and $\Phi\Psi = 1_{\mathbb{W}}^{2\delta}$.*

Although it does not define a metric on the space of persistence modules, the notion of closeness between two persistence module may be defined as the smallest non negative δ such that they are δ-interleaved. Moreover, it allows to formalize the following fundamental theorem Chazal et al. (2009a, 2016a).

Theorem 7 (Stability of persistence)**.** *Let $\mathbb{V}$ and $\mathbb{W}$ be two q-tame persistence modules. If $\mathbb{V}$ and $\mathbb{W}$ are δ-interleaved for some $\delta \geq 0$, then*

$$\mathrm{d_b}(\mathrm{dgm}(\mathbb{V}), \mathrm{dgm}(\mathbb{W})) \leq \delta.$$

Although purely algebraic and rather abstract, this result is a efficient tool to easily establish concrete stability results in TDA. For example we can easily recover the first persistence stability result that appeared in the literature (Cohen-Steiner et al., 2005).

Theorem 8. *Let $f, g : M \to \mathbb{R}$ be two real-valued functions defined on a topological space M that are q-tame, i.e. such that the sublevel sets filtrations of f and g induce q-tame modules at the homology level. Then for any integer k,*

$$\mathrm{d_b}(\mathrm{dgm}_k(f), \mathrm{dgm}_k(g)) \leq \|f - g\|_\infty = \sup_{x \in M} |f(x) - g(x)|$$

where $\mathrm{dgm}_k(f)$ *(resp.* $\mathrm{dgm}_k(g)$*) is the persistence diagram of the persistence module* $(H_k(f^{-1}(-\infty, r]))|r \in \mathbb{R})$ *(resp.* $(H_k(g^{-1}(-\infty, r]))|r \in \mathbb{R})$*) where the linear maps are the one induced by the canonical inclusion maps between sublevel sets.*

Proof. Denoting $\delta = \|f - g\|_\infty$ we have that for any $r \in \mathbb{R}$, $f^{-1}(-\infty, r]) \subseteq g^{-1}(-\infty, r + \delta])$ and $g^{-1}(-\infty, r]) \subseteq f^{-1}(-\infty, r + \delta])$. This interleaving between the sublevel sets of f induces a δ-interleaving between the persistence modules at the homology level and the result follows from the direct application of Theorem 7. □

Theorem 7 also implies a stability result for the persistence diagrams of filtrations built on top of data.

Theorem 9. *Let $\mathbb{X}$ and $\mathbb{Y}$ be two compact metric spaces and let* $\mathrm{Filt}(\mathbb{X})$ *and* $\mathrm{Filt}(\mathbb{Y})$ *be the Vietoris-Rips of Čech filtrations built on top $\mathbb{X}$ and $\mathbb{Y}$. Then*

$$d_b\left(\mathrm{dgm}(\mathrm{Filt}(\mathbb{X})), \mathrm{dgm}(\mathrm{Filt}(\mathbb{Y}))\right) \leq 2 d_{GH}(\mathbb{X}, \mathbb{Y})$$

where $\mathrm{dgm}(\mathrm{Filt}(\mathbb{X}))$ *and* $\mathrm{dgm}(\mathrm{Filt}(\mathbb{Y}))$ *denote the persistence diagram of the filtrations* $\mathrm{Filt}(\mathbb{X})$ *and* $\mathrm{Filt}(\mathbb{X})$*.*

As we already noticed in the Example 3 of Section 5.2, the persistence diagrams can be interpreted as multiscale topological features of $\mathbb{X}$ and $\mathbb{Y}$. In addition, Theorem 9 tells us that these features are robust with respect to perturbations of the data in the Gromov-Hausdorff metric. They can be used as discriminative features for classification or other tasks - see, for example, Chazal et al. (2009b) for an application to non rigid 3D shapes classification.

We now give similar results for the alternative persistence homology representations introduced before. From the definition of persistence landscape we immediately observe that $\lambda(k, \cdot)$ is one-Lipschitz and thus similar stability properties are satisfied for the landscapes as for persistence diagrams.

Proposition 1 (Stability of persistence landscapes Bubenik (2015).)**.** *Let* dgm *and* dgm$'$ *be two persistence diagrams (without their essential parts). For any $t \in \mathbb{R}$ and any $k \in \mathbb{N}$, we have:*
(i) $\lambda(k, t) \geq \lambda(k + 1, t) \geq 0$.
(ii) $|\lambda(k, t) - \lambda'(k, t)| \leq \mathrm{d_b}(\mathrm{dgm}, \mathrm{dgm}')$.

A large class of linear representations are continuous with respect to the Wasserstein metric W_s in the space of persistence diagrams and with respect to the Banach norm of the linear representation of persistence. Generally speaking, it is not always possible to upper bound the modulus of continuity of the linear representation operator. However, in the case where $s = 1$ it is even possible to show a stability result if the weight function take small values for points close to the diagonal, see Divol and Lacombe (2020); Hofer et al. (2019b).

Stability versus discriminative capacity of persistence representations. The results of Divol and Lacombe (2020) show that continuity and stability is only possible with weigh functions taking small values for points close to the diagonal. However, in general there is no specific reason to consider that points close to the diagonal are less important than others, given a learning task. In a machine learning perspective, it is also relevant to design linear representation with general weigh functions, although it would be more difficult to prove consistency of the